

HEMATOPOIESIS AND STEM CELLS

IFN- γ drives long-term bone marrow niche dysfunction following doxorubicin-based chemotherapy

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KEY POINTS

- DOX remodels the BM niche, inducing arteriolar damage, MSC dysfunction, and sustained expansion of phenotypic HSCs.
- IFN- γ signaling from CD8⁺ T cells promotes chronic inflammation and reprograms stromal and vascular compartments in the BM niche.

Cancer survivors experience long-term skeletal and hematopoietic complications that limit quality of life following chemotherapy (CTX), yet the mechanisms underlying these defects remain incompletely understood. In this study, we used a murine model of doxorubicin (DOX)-based leukemia induction therapy to show that CTX induces inflammatory remodeling of the bone marrow (BM) niche. DOX treatment resulted in a loss of arteriolar vasculature, blockade of mesenchymal stromal cell (MSC) differentiation, trabecular bone loss, and reduced niche capacity to maintain hematopoietic stem cells. These defects were accompanied by aberrant immune activation within the BM, marked by increased interferon gamma (IFN- γ) production by CD8⁺ T cells. Inhibition of IFN- γ signaling partially restored arteriolar vessels and adipogenic differentiation. Furthermore, combined IFN- γ blockade with deferoxamine mesylate, which promotes vascular recovery, attenuated CTX-associated skeletal damage. Consistent with these findings, paired BM samples collected at diagnosis and after CTX from patients with leukemia exhibited altered MSC lineage priming, upregulation of inflammatory pathways, and expansion of BM CD8⁺ memory T cells after treatment. Together, these findings implicate IFN- γ -driven chronic inflammatory remodeling as a central mechanism of CTX-associated BM niche dysfunction and pinpoint inflammatory signaling as a potential target to preserve BM function and long-term tissue integrity.

Introduction

Chemotherapeutic agents have remained a cornerstone of cancer care since the 1960s, significantly extending survival in both hematologic and solid tumors. However, their nonspecific

cytotoxicity extends far beyond malignant cells, inflicting collateral damage on healthy tissues. Cancer survivors experience long-term skeletal complications, impaired steady-state hematopoiesis, and an increased risk of developing secondary malignancies.¹⁻³ Among these agents, doxorubicin (DOX),

a widely used anthracycline for the treatment of leukemia and lymphoma, has been well documented to be associated with hematopoietic and skeletal toxicities. DOX exerts antitumor effects by inhibiting topoisomerase II, thereby inducing DNA double-strand breaks and apoptosis.⁴ Despite its clinical utility, the mechanisms by which DOX remodels the bone marrow (BM) microenvironment remain poorly defined. This knowledge gap hampers the development of strategies to preserve BM tissue integrity and long-term hematopoietic support in patients receiving chemotherapy (CTX). Therefore, a deeper understanding of how CTX disrupts the BM niche is essential to mitigate its long-term complications in cancer survivors.

Hematopoiesis is maintained by a rare pool of hematopoietic stem cells (HSCs) that reside within the BM niche. This specialized microenvironment consists of diverse and highly organized cellular components, including leptin receptor-positive (Lepr⁺) mesenchymal stromal cells (MSCs) and vascular endothelial cells (ECs), which provide essential signals that regulate HSC quiescence, self-renewal, and lineage output.⁵ In addition to their hematopoietic supportive functions, Lepr⁺ MSCs serve as the principal skeletal progenitors in adult mice, sustaining the production of osteoblasts and adipocytes.⁶ Although the BM niche exhibits regenerative capacity in response to stress, the long-term effects of CTX on niche biology remain unclear. In this study, combining detailed characterization of DOX-mediated remodeling of the murine mouse BM microenvironment with validation in BM aspirate samples from CTX-treated patients, we showed that CTX drives inflammatory remodeling of the BM microenvironment, positioning inflammation as a major mechanism linking CTX exposure to long-term BM niche dysfunction.

Methods

Animals

All mouse experiments were performed in accordance with protocols approved by the Animal Resources Centre at the University Health Network (UHN). For DOX treatment, mice were administered either a single dose (5 or 10 mg/kg) or 4 weekly doses (5 mg/kg) through retro-orbital injections.

Patient samples

All samples were collected with informed consent in accordance with procedures approved by the research ethics board (REB) of the UHN (REB 20-6086). Cryopreserved paired BM aspirates collected from patients at the time of leukemia diagnosis and after CTX were obtained from the Princess Margaret Leukemia Biobank (supplemental Table 1, available on the *Blood* website).

Detailed methods for all functional studies and computational analyses are provided in supplemental Methods.

Results

DOX treatment compromises the hematopoietic supportive capacity of BM

To evaluate the impact of DOX on the BM, we administered 4 weekly doses of DOX through IV injection to C57BL/6 mice (aged 8-10 weeks), a regimen designed to mimic the induction

phase in the treatment of acute lymphoblastic leukemia (ALL; Figure 1A). Following an initial drop, total BM cellularity returned to baseline by 6 days after treatment (Figure 1B; supplemental Figure 1A). Flow cytometry revealed a marked reduction in lymphoid populations, including B220⁺ B cells and CD3⁺ T cells, with an increase in CD11b⁺Gr1⁺ myeloid cells (Figure 1C-D; supplemental Figure 1B). Notably, these changes were resolved by 8 weeks, indicating no lasting effect on mature hematopoietic output.

We next assessed hematopoietic stem and progenitor cell (HSPC) dynamics in response to DOX treatment (supplemental Figure 1C). We observed pronounced expansion of cKit⁺ HSPCs 6 days after treatment (Figure 1E-F; supplemental Figure 1D). Within HSPCs, we observed a decrease in the frequency of lymphoid-primed multipotent progenitor 4 (MPP4), whereas relative proportions of HSC, MPP2, and MPP3 remained unchanged. Increased numbers of HSPC populations and myeloid skewing indicated induction of emergency hematopoiesis in response to DOX-mediated hematopoietic injury. Although mature lineages returned to baseline by 8 weeks (Figure 1D; supplemental Figure 1B), HSPC and HSC numbers remained elevated in DOX-treated mice (Figure 1F), suggesting a sustained loss of HSC quiescence.

To determine whether DOX induces cell-intrinsic defects in HSCs, we isolated HSCs from control or DOX-treated mice 8 weeks after treatment and transplanted them into lethally irradiated recipients (supplemental Figure 1E). Although HSCs from DOX-treated donors showed a transient delay in B-cell reconstitution at 8 weeks after transplant, overall donor chimerism was comparable between groups (supplemental Figure 1F). Consistently, we observed no reduction in donor engraftment (supplemental Figure 1G) or HSC abundance (supplemental Figure 1H) in the BM at 16 weeks after transplantation. Collectively, these data indicate that DOX treatment does not induce major cell-intrinsic alterations in HSCs. We then assessed whether DOX impairs the hematopoietic supportive function of the BM niche. To test this, we treated recipient mice with DOX and transplanted them with healthy HSCs (Figure 1G). All mice survived up to 16 weeks after transplant, with similar donor chimerism and lineage output between control and DOX-treated hosts (Figure 1H; supplemental Figure 1I-J). However, DOX-treated recipients showed a significant reduction in HSC frequency and number at 16 weeks (Figure 1I), indicating that although initial HSC engraftment is supported, prior DOX exposure induces durable remodeling of the BM niche that impairs long-term HSC maintenance. These findings align with clinical observations that despite normalized blood counts, patients with leukemia who were treated with high-dose CTX have reduced hematopoietic progenitors years after transplantation.^{7,8}

DOX disrupts niche composition and reprograms niche cell states toward inflammation

To define transcriptional changes underlying the observed BM niche dysfunction, we performed single-cell RNA sequencing (scRNA-seq) on nonhematopoietic (CD45⁻lineage⁻cKit⁻) BM cells 8 weeks after DOX treatment (Figure 2A). Unsupervised clustering of 20 616 quality-filtered cells identified 18 distinct clusters, annotated using our previously published BM niche

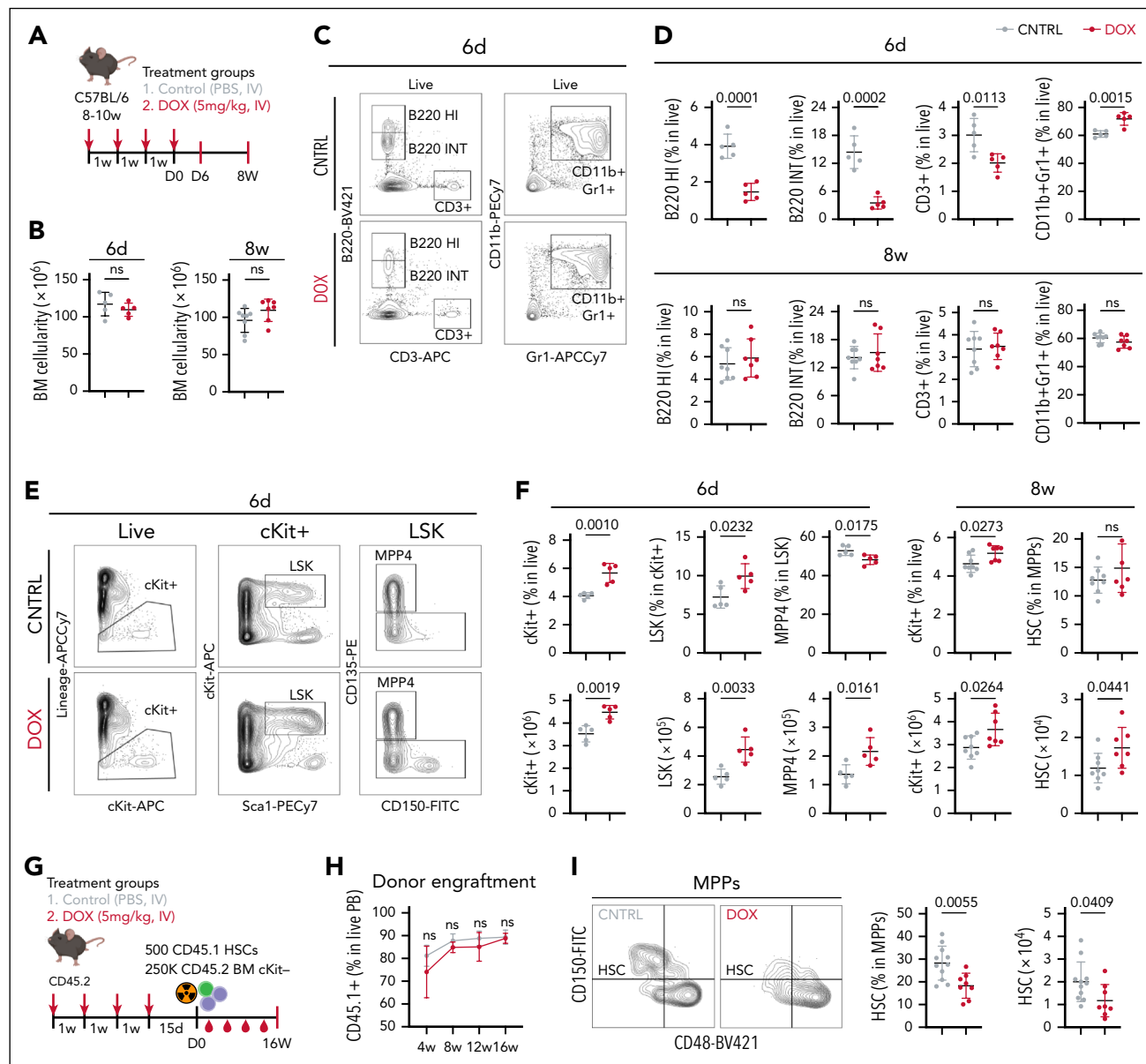


Figure 1. DOX treatment impairs the hematopoietic supportive capacity of the BM niche. (A) Schematic of serial control and DOX treatment (4 doses, 5 mg/kg) regimen. (B) Total BM cellularity of control and DOX-treated animals 6 days (6d) and 8 weeks (8w) after treatment. $n = 5$ to 8; unpaired t test. (C) Representative images of flow cytometry analysis on mature B cells (B220^{HI}), B lineage progenitors (B220^{INT}), T cells (CD3⁺), and myeloid cells (Gr1⁺CD11b⁺) at 6 days after control and DOX treatments. (D) Relative frequencies of mature B cells, B lineage progenitors, T cells, and myeloid cells at 6 days and 8 weeks after treatment. $n = 5$ to 8; unpaired t test. (E) Representative images of flow cytometry analysis of HSPCs at 6 days after treatment. (F) Relative frequencies and numbers of cKit⁺ HSPCs, LSK (lineage cKit⁺Sca1⁺), and MPP4 (LSK⁺CD135⁺CD150⁺) populations at 6 days and 8 weeks after treatment. $n = 5$ to 8; unpaired t test. (G) Schematic of HSC transplantation into recipient mice treated with control or DOX treatment. (H) Donor HSC-derived engraftment in the peripheral blood of control and DOX-treated recipients at different time points. $n = 8$ to 11; multiple unpaired t tests with false discovery rate (FDR) correction. (I) Analysis of HSCs in DOX-treated recipients 16 weeks after transplant. Representative images of flow cytometry analysis (left). Relative frequency and number of HSCs (LSK⁺CD135⁺CD48⁺CD150⁺) in control and DOX-treated host BM (right). $n = 8$ to 11; unpaired t test. CNTRL, control; LSK, lineage cKit⁺Sca1⁺; MPPs, multipotent progenitors; ns, not significant; PB, peripheral blood; PBS, phosphate-buffered saline. Created with BioRender.com. Li, X. (2026) <https://BioRender.com/zdi6tp4>.

atlas.⁹ This approach captured main stromal and vascular populations in the BM, including chondrocytes, vascular ECs, fibroblasts, MSCs, myofibroblasts, osteoblasts, pericytes, Schwann cells, and smooth muscle cells (SMCs; Figure 2B; supplemental Figure 2A). MSCs resolved into 2 subsets with distinct lineage priming, which were MSC-Adipo and MSC-Osteo. ECs segregated into type R,¹⁰ arterial/arteriolar, transitional/type H,¹¹ and sinusoidal subsets (supplemental Figure 2B).

We first examined whether DOX alters BM niche composition (Figure 2C; supplemental Figure 2C). Our analysis revealed an expansion of adipogenic-primed MSCs and a reduction in osteogenic-primed MSCs in DOX-treated animals (Figure 2D). Within the vascular compartment, we observed a loss of arteriolar and transitional ECs. Furthermore, pericytes and SMCs, the supporting cells adjacent to arterial/arteriolar vessels, were reduced after DOX treatment (supplemental Figure 2C). We next validated the compositional changes in BM vasculature

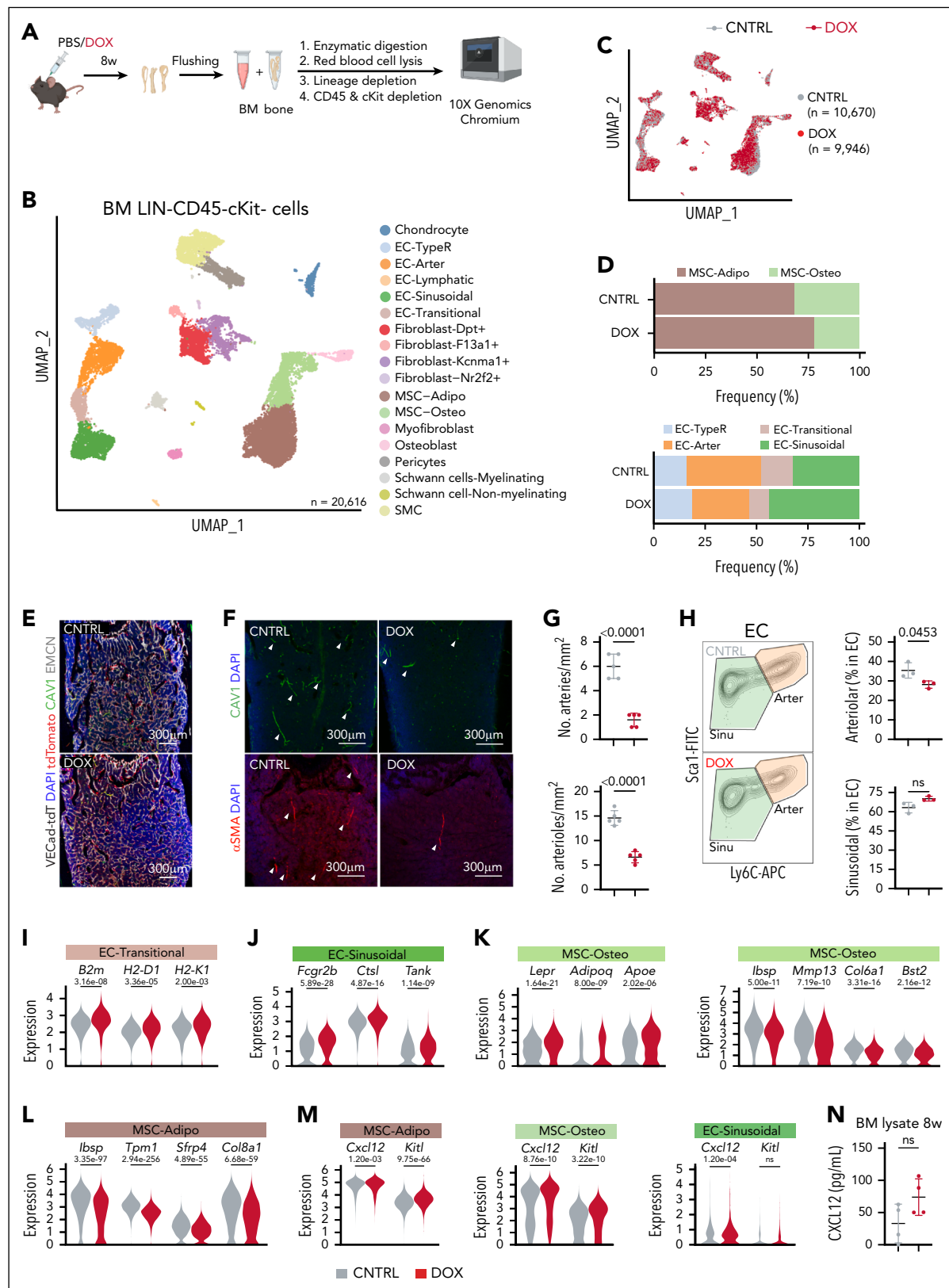


Figure 2. Changes in the transcriptional heterogeneity of the BM microenvironment after DOX treatment. (A) Sample preparation workflow of scRNA-seq on BM non-HSCs at 8 weeks (8w) after control and DOX treatment. (B) Uniform manifold approximation and projection (UMAP) plot showing different clusters identified in the BM niche data set, combining control and DOX data sets. (C) Distribution of cells isolated from control and DOX-treated animals on the UMAP plot. (D) Composition changes in the BM stromal and EC compartment. Frequencies of MSC-Adipo and MSC-Osteo in all MSCs (top). Frequencies of EC-type R, arterial/arteriolar, transitional, and sinusoidal in all ECs (bottom). (E) Representative confocal images of BM vasculature in femurs of control or DOX-treated mice at 8 weeks after treatment, male. (F) Representative confocal images showing CAV1⁺ arterial/arteriolar vessels and α SMA⁺ pericytes in femurs of DOX-treated animals at 8 weeks after treatment, female. (G) Number of arteries

using endothelial reporter VECad-Cre;tdTomato (VECad-tdT) mice. Flow cytometry analysis (supplemental Figure 2D) and confocal imaging (Figure 2E; supplemental Figure 2E) showed no marked reduction in the overall abundance of tdT⁺ ECs in DOX-treated BM. However, consistent with scRNA-seq results, we found a pronounced decrease in caveolin 1-expressing (CAV1⁺) arterial/arteriolar vessels and alpha-smooth muscle actin-positive (α SMA⁺) pericytes/SMCs 8 weeks after DOX treatment (Figure 2F-G). Similarly, flow cytometry analysis showed a significant decrease in arteriolar ECs (Ly6C^{HIGH}-ca1^{HIGH}; Figure 2H). Arteriolar ECs have been reported as a critical vascular subset in supporting hematopoiesis, especially in maintaining quiescent HSCs.^{12,13} Thus, DOX may compromise the arteriolar niche, impairing the ability of BM to support HSC quiescence. Together, these results indicate that DOX induces persistent alterations in the BM niche composition that extend well beyond the treatment window.

We next performed differential gene expression (DGE) analysis across BM niche clusters between treatment groups. Within ECs, transitional and sinusoidal subsets upregulated genes associated with inflammatory responses (Figure 2I-J), whereas Type R, arterial/arteriolar, and transitional subsets increased expression of angiogenesis-related genes (supplemental Figure 2F), suggesting activation of repair programs in response to DOX-induced vascular injury. In MSCs, DGE results revealed an adipogenic lineage skewing, where MSC-Osteo showed an upregulation of adipogenic markers and downregulation of osteogenic genes (Figure 2K), with a similar osteogenic downregulation in MSC-Adipo (Figure 2L). These findings indicate inflammatory niche remodeling accompanied by MSC fate skewing toward a more adipogenic-primed state. Notably, both MSC subsets showed increased expression of *Cxcl12* and *Kitl*, 2 key niche-derived factors that regulate HSC retention and maintenance^{14,15} (Figure 2M). Elevated *Cxcl12* expression was also detected in sinusoidal ECs at 8 weeks after treatment. Consistently, multiplex cytokine profiling confirmed elevated CXCL12 protein levels in BM lysates (Figure 2N), compatible with a compensatory niche response that remains insufficient to fully restore normal HSC support. Together, these data demonstrate that DOX induces persistent, cell type-specific shifts in the BM niche transcriptome, characterized by inflammatory activation and impaired MSC differentiation programs.

DOX treatment impairs MSC function

Given the DOX-mediated transcriptional and compositional changes observed in MSCs, we next examined the impact of DOX on MSC stem cell function. To this end, we used Lepr-Cre;tdTomato (Lepr-tdT) mice to label MSCs and their progeny and tracked tdT⁺ cells after DOX treatment (supplemental Figure 3A). MSC frequency remained unchanged at early time points (at 6 days, 4 weeks, and 8 weeks) after treatment. However, by 20 weeks, we observed a striking increase in MSC abundance, as detected by both flow cytometry and imaging (Figure 3A-B; supplemental Figure 3C), despite no change in

total BM cellularity (supplemental Figure 3B). Cell cycle analysis showed that >90% of MSCs remained in G0/G1 phase at 15 days, 8 weeks, and 20 weeks after treatment (supplemental Figure 3D-G), suggesting that the increased MSC abundance was not because of increased proliferation but rather a potential block in their differentiation capacity.

To assess whether DOX impairs the clonogenic potential of MSCs, we performed fibroblast colony-forming unit assays on BM from control and DOX-treated Lepr-tdT mice at 8 weeks after treatment. DOX-treated samples showed a significant reduction in tdT⁺ colony number and size (Figure 3C-E), indicating loss of clonogenic MSCs and diminished proliferative capacity. We next examined whether DOX affects MSC lineage commitment. Based on the altered MSC subset composition observed in our scRNA-seq analysis (Figure 2D), we quantified changes in the proportion of osteogenic-primed (CD200^{HIGH}CD63^{HIGH}) and adipogenic-primed (CD200^{LOW}CD63^{LOW})¹⁶ MSCs with flow cytometry. No significant differences were detected at 4 weeks (supplemental Figure 3H); however, by 8 weeks and 20 weeks, DOX-treated mice showed a shift toward adipogenic-primed MSCs and a decrease in the osteogenic-primed fraction (Figure 3F-G). Consistently, ex vivo differentiation assays showed reduced osteogenic potential and intact adipogenesis in MSCs from DOX-treated mice (supplemental Figure 3I-J), indicating impaired skeletal differentiation capacity.

To quantify MSC osteogenic output in vivo, we crossed Lepr-tdT mice with Col1a1-GFP osteoblast reporters (Lepr-tdT;Col1a1-GFP), labeling MSC-derived osteoblasts with both tdT and green fluorescent protein (GFP). At 8 weeks after treatment, confocal imaging and colocalization analysis showed a significant reduction in tdT⁺GFP⁺ osteoblasts in DOX-treated BM (Figure 3H-J). This deficit persisted at 20 weeks (supplemental Figure 3K), indicating a sustained impairment in MSC osteogenesis. Surprisingly, we also identified loss of tdT⁺Perilipin⁺ mature adipocytes at both 8 and 20 weeks after treatment (supplemental Figure 3L; Figure 3K-L). Despite transcriptional priming of adipogenic programs in MSCs (Figure 2K-L) and intact ex vivo adipogenic potential (supplemental Figure 3J), adipogenic output was not increased in vivo, suggesting that DOX-induced changes in the tissue microenvironment may limit terminal adipocyte differentiation and/or maintenance. Together, these results indicate that DOX disrupts MSC multilineage differentiation, leading to long-term deficits in both osteoblast and adipocyte production.

To assess whether DOX-induced MSC defects affect skeletal integrity, we performed microcomputed tomography on femurs 8 weeks after treatment. DOX-treated mice showed reduced femoral length, decreased bone mineral density (BMD), and pronounced trabecular bone loss (Figure 4A-B). Histological analysis confirmed a narrowed tibial growth plate (Figure 4C; supplemental Figure 4A), consistent with impaired longitudinal growth. Despite modest weight loss in males,

Figure 2 (continued) and arterioles in control and DOX-treated animals, quantified based on panel F. n = 5; unpaired t test. (H) Changes in vascular composition following DOX treatment. Representative flow cytometry analysis of vascular subset (left). Relative frequencies of vascular subsets (right). n = 3; unpaired t test. (I-J) Expression of inflammatory genes in EC-transitional (I) and EC-Sinusoidal (J). (K-L) Expression of genes associated with adipogenesis and osteogenesis in MSC-Osteo (K) and MSC-Adipo (L). (M) Expression of *Cxcl12* and *Kitl* in different niche subsets. Mann-Whitney U test with FDR correction for panels I-M. (N) CXCL12 protein level, measured by Olink cytokine assay, in BM lysate at 8 weeks after treatment, n = 4; unpaired t test. Arter, arterial/arteriolar; CNTRL, control; ns, not significant; PBS, phosphate-buffered saline. Created with BioRender.com. Li, X. (2026) <https://BioRender.com/zdi6tp4>.

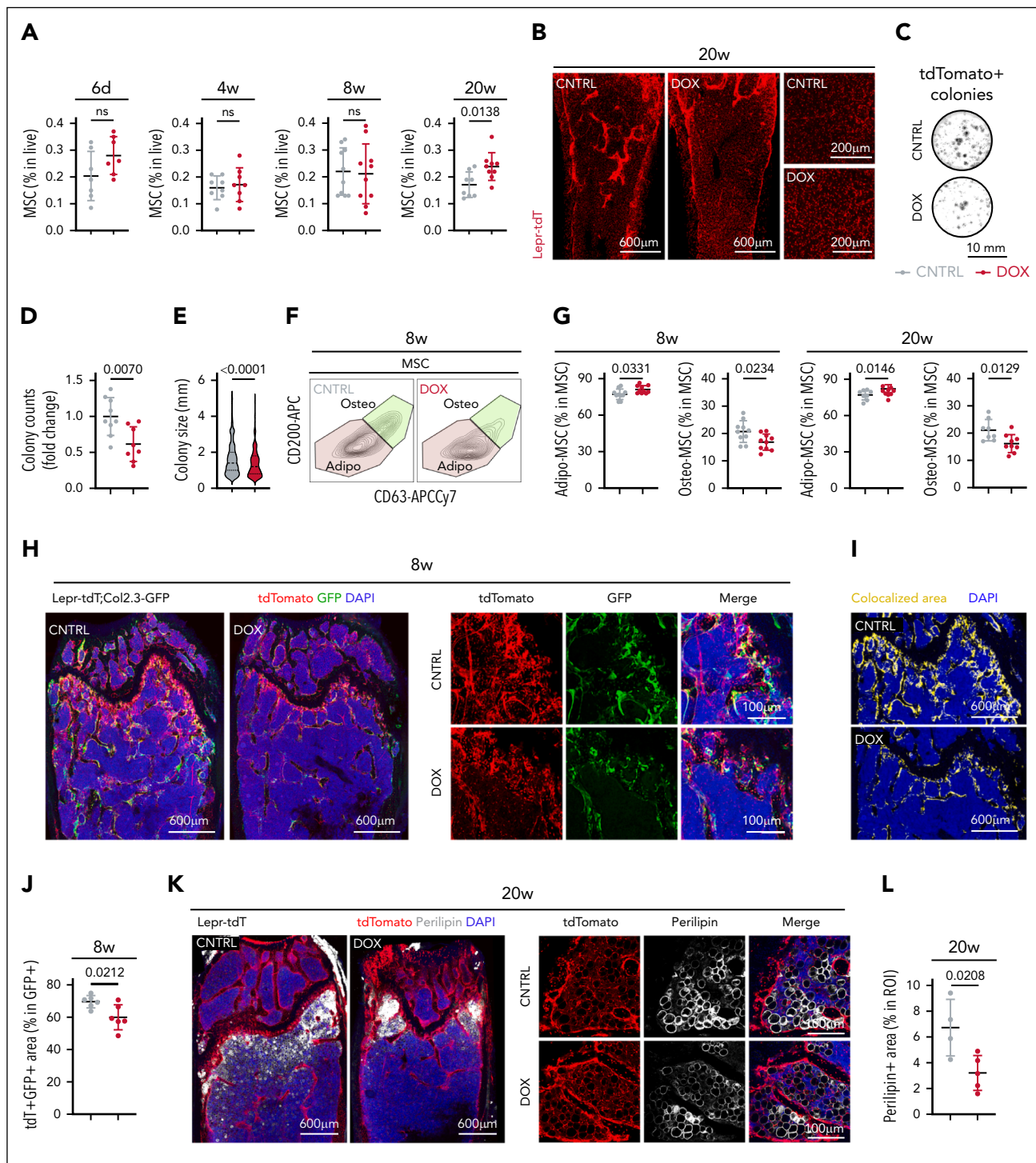


Figure 3. DOX treatment impairs MSC function. (A) Relative frequency of tdT⁺ MSCs at 6 days (6d), 4 weeks (4w), 8 weeks (8w), and 20 weeks (20w) after treatment. *n* = 7 to 10; unpaired *t* test. (B) Representative confocal images showing accumulation of tdT⁺ MSC in DOX-treated BM at 20 weeks after treatment; female. (C) Representative images of MSC-derived tdT⁺ colonies from 750 000 unlysed total BM cells from control and DOX-treated animals. (D) Number of tdT⁺ colonies, normalized to control. *n* = 8 to 9; unpaired *t* test. (E) Diameters of all tdT⁺ colonies formed in control and DOX BM. Mann-Whitney *U* test. (F) Representative images of flow cytometry analysis of MSC subset composition at 8 weeks after treatment. (G) Relative frequencies of adipogenic-primed and osteogenic-primed MSCs at 8 weeks and 20 weeks after control and DOX treatment. *n* = 8 to 10; unpaired *t* test. (H) Representative confocal images of MSC osteogenic lineage tracing in femurs at 8 weeks after control and DOX treatment, male. (I) Colocalized area of GFP and tdT signal of confocal images in panel H. (J) Fraction of GFP⁺ area overlapped with tdTomato signal, indicating the proportion of MSC-derived osteoblasts (GFP⁺tdT⁺) in total osteoblasts (GFP⁺) at 8 weeks after control and DOX treatment, quantified based on panel H. *n* = 6; unpaired *t* test. (K) Representative confocal images of MSC adipogenic lineage tracing in femurs at 20 weeks after treatment, female. (L) Proportion of perilipin⁺ area in the metaphyseal region of interest, quantified based on panel K. *n* = 4 to 5; unpaired *t* test. CNTRL, control; DAPI, 4',6-diamidino-2-phenylindole; ns, not significant; ROI, region of interest.

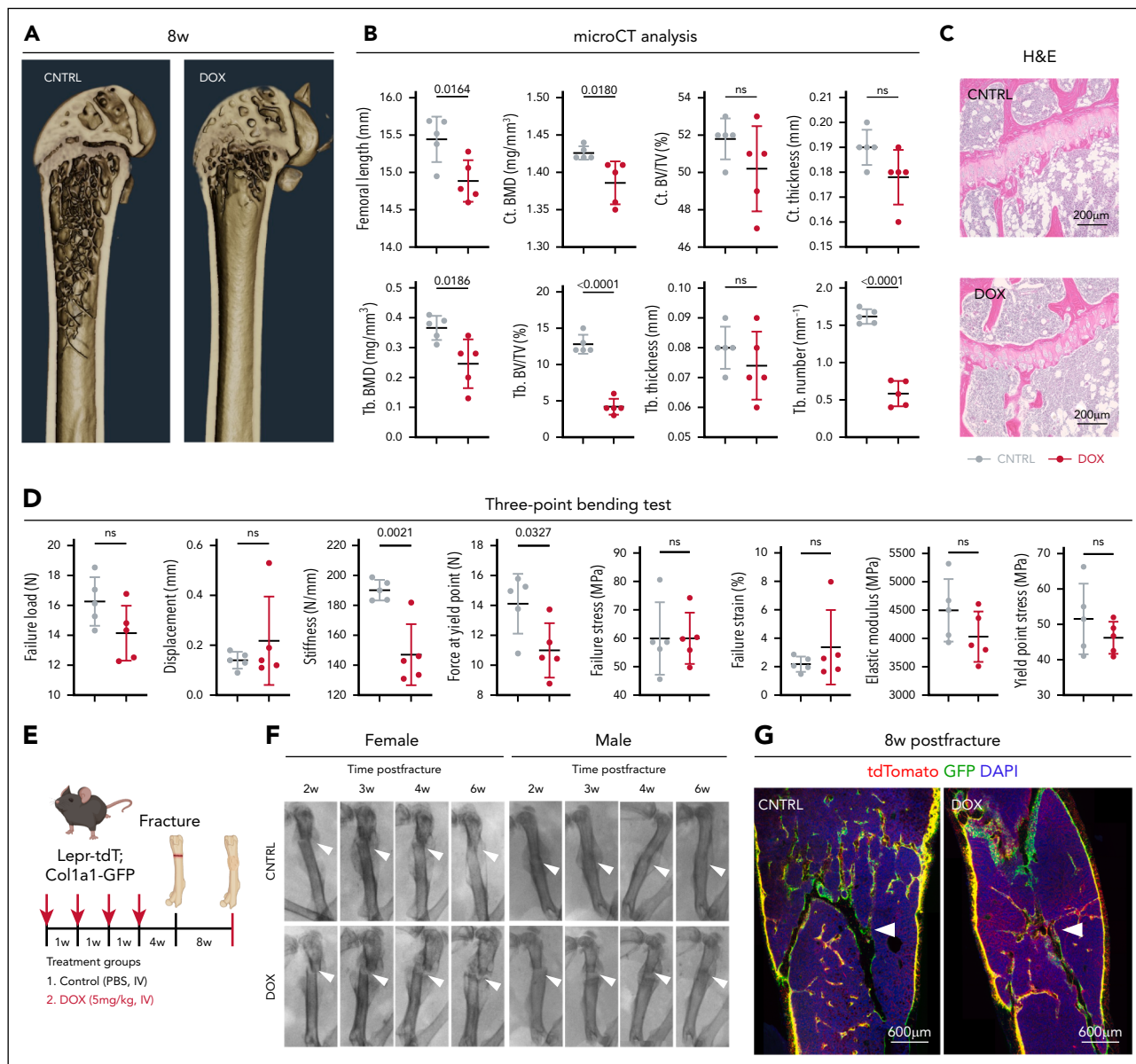


Figure 4. DOX treatment leads to long-term skeletal defects. (A) Representative reconstructed micro-CT images of femurs at 8 weeks (8w) after treatment, female. (B) Stereological parameters measured in micro-CT analysis on control and DOX-treated femurs. $n = 5$, female, unpaired t test. (C) Representative image of hematoxylin and eosin stain on tibias of control and DOX-treated mice at 8 weeks after treatment, female. (D) Results of 3-point bending tests on diaphyseal cortical bone from femurs of control and DOX-treated animals 8 weeks after treatment. $n = 5$, female; unpaired t test. (E) Schematic of bicortical fracture experiment on DOX-treated mice. (F) Representative serial CT scans of fractured femurs from control and DOX-treated mice at the indicated time points after surgery. (G) Representative confocal images of regenerated femurs at 8 weeks after fracture, female. White arrows indicate the fracture site. BV/TV, bone volume/tissue volume; BMD, bone mineral density; BV, bone volume; Ct, cortical; CT, computed tomography; CNTRL, control; H&E, hematoxylin and eosin stain; micro-CT, microcomputed tomography; MPa, megapascal; N, newton; ns, not significant; Tb, trabecular; TV, tissue volume. Created in [BioRender.com](https://BioRender.com/zdi6tp4). Li, X. (2026) <https://BioRender.com/zdi6tp4>.

overall body size and activity were unaffected in DOX-treated mice (supplemental Figure 4B-C). Three-point bending tests revealed reduced femoral stiffness and strength (Figure 4D). As failure stress and elastic modulus remained unchanged, these compromised biomechanical properties are attributable to altered cortical geometry rather than changes in bone matrix properties. Notably, DOX-mediated bone loss persisted up to 20 weeks (supplemental Figure 4D), and even a single DOX injection was sufficient to reduce BMD and bone mass by 8 weeks (supplemental Figure 4E-F). These findings align with clinical reports of height deficit, low BMD, and increased fracture risk in DOX-treated patients.^{3,17,18} Combined with our

lineage-tracing data, these results suggest that DOX-induced skeletal defects are driven, at least in part, by impaired MSC osteogenesis.

In addition to maintaining bone integrity at steady state, $Lepr^+$ MSCs play a critical role in bone regeneration. To test whether DOX impairs MSC regenerative capacity, we induced femoral fracture in $Lepr$ -tdT;Col1a1-GFP mice 4 weeks after treatment and examined the kinetics of bone formation during fracture healing (Figure 4E). Serial computed tomography scans revealed a delay in callus calcification at the fracture site in DOX-treated mice compared with controls (Figure 4F). Notably,

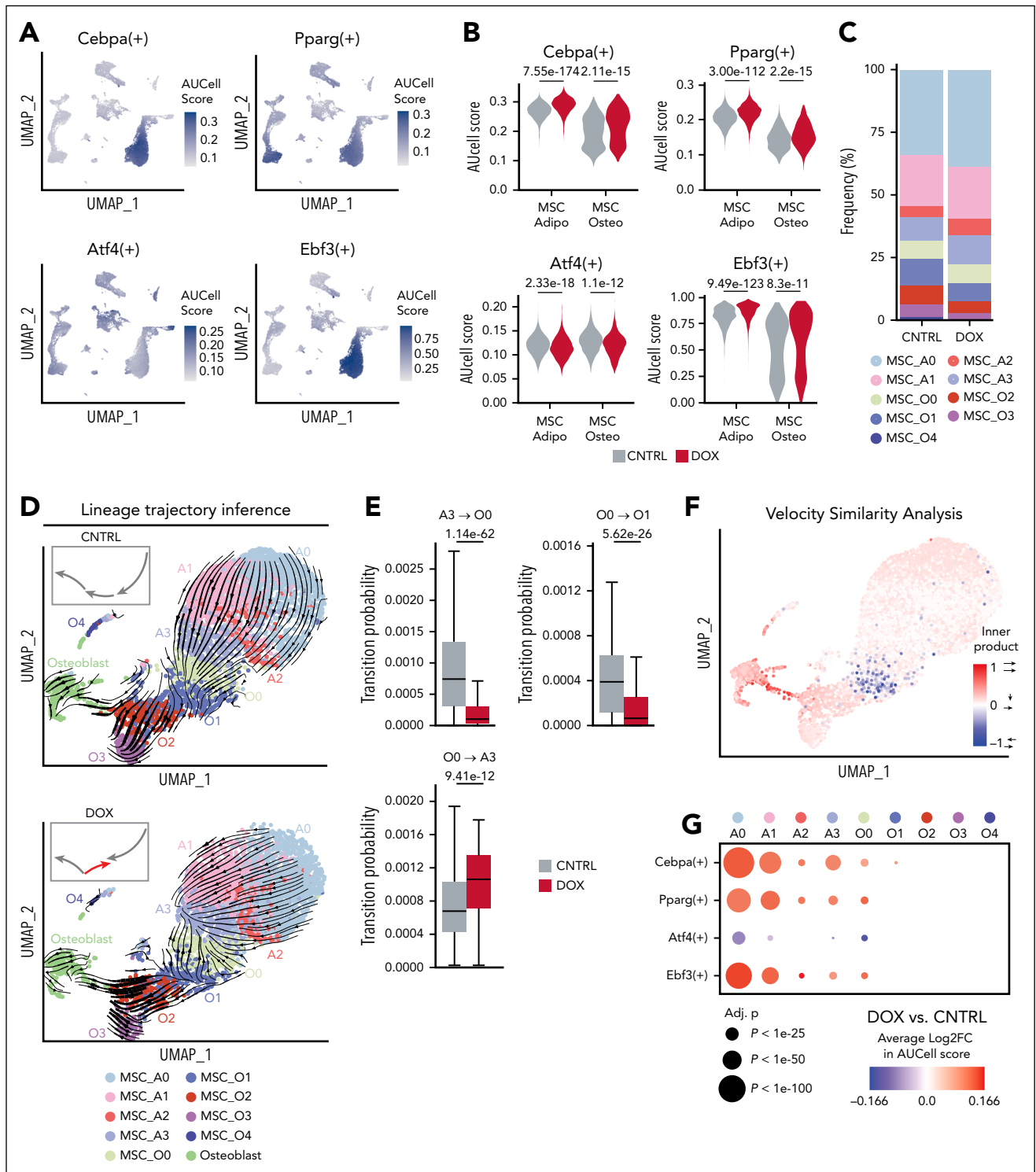


Figure 5. Lineage trajectory inference reveals impaired MSC osteogenesis. (A) Predicted activities of indicated TFs, shown as AUCell scores, displayed on the UMAP of the BM niche data set. (B) Difference in predicted activity of indicated TFs in MSC-Adipo and MSC-Osteo clusters from control and DOX-treated mice. Mann-Whitney *U* test with FDR correction. (C) Frequencies of MSC-Adipo and MSC-Osteo subclusters in all MSCs per treatment condition. (D) Velocities visualized as vector streamlines on UMAP plots composed of MSCs and osteoblasts in the control and DOX data set. (E) Transition probabilities between indicated MSC-Adipo and MSC-Osteo subclusters from control and DOX data set, derived from RNA velocity analysis. Mann-Whitney *U* test with FDR correction. (F) Velocity similarity, calculated as the inner product of velocities of cells from 2 treatment conditions, on the UMAP consisting of MSCs and osteoblasts. A number close to 1 indicates velocities have similar directions (ie, similar lineage trajectory); a number close to -1 indicates opposite velocity directions. (G) Differential expression analysis of predicted TF activities in MSC-Adipo and MSC-Osteo subclusters comparing control and DOX data sets. Mann-Whitney *U* test with FDR correction. Adj. p, adjusted probability value; CNTRL, control; Log2FC, Log₂ fold change.

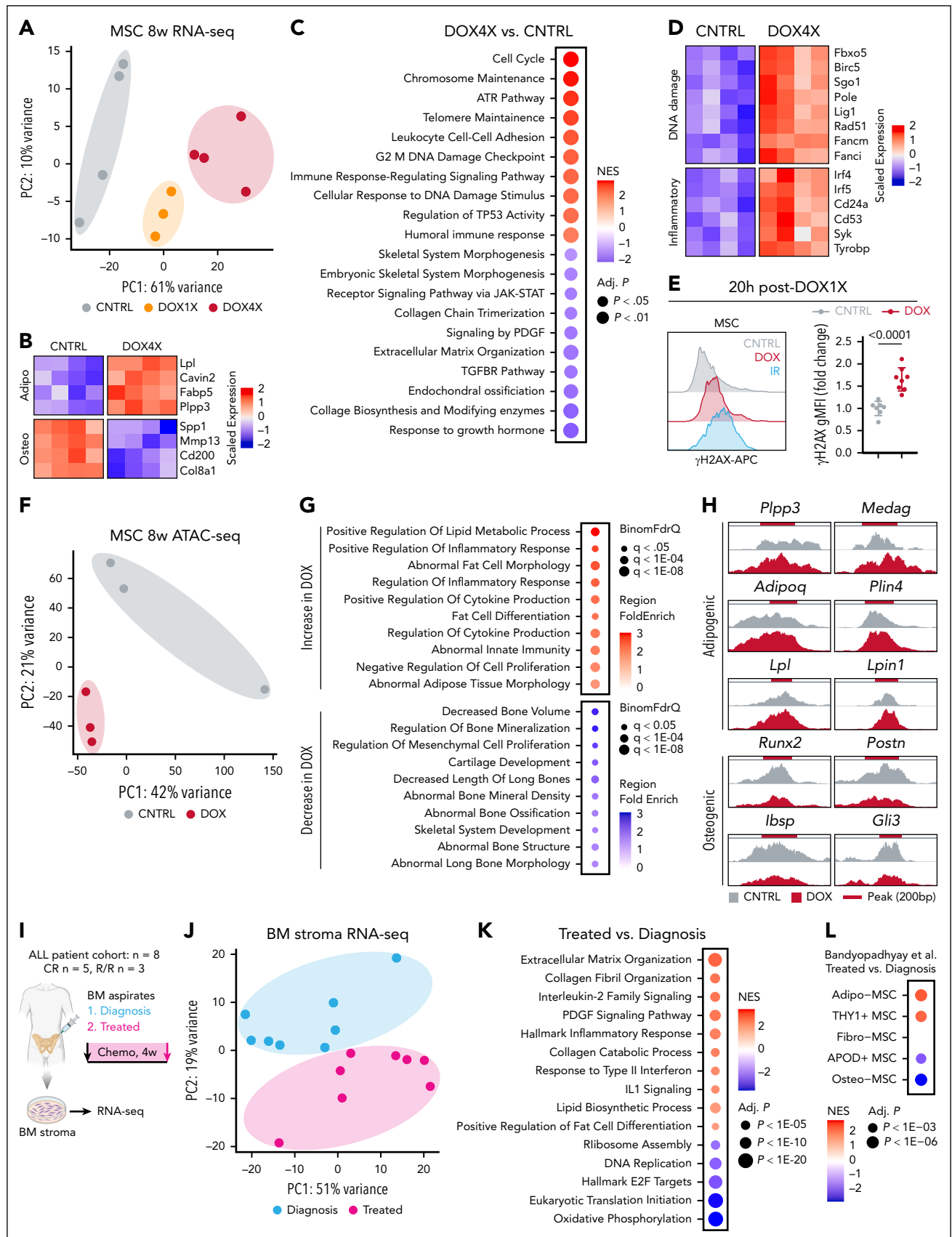


Figure 6. Epigenetic and transcriptional dysregulation drives impaired MSC osteogenesis. (A) Principal component analysis (PCA) plot of RNA-seq data set of MSCs isolated from mice receiving control, single DOX (10 mg/kg), and serial DOX (5 mg/kg) treatment 8 weeks (8w) later. $n = 3$ to 4. (B) Expression of differentially expressed markers for adipogenic-primed and osteogenic-primed MSCs in bulk MSC RNA-seq following control and DOX4X treatment. Wald test with FDR correction. (C) Gene set

confocal imaging of fractured femurs at 8 weeks after injury showed that osteoblasts lining the original cortical bone were GFP⁺tdT⁺, whereas osteoblasts lining the newly formed bone tissue at the injury site were derived from MSCs (GFP⁺tdT⁺), suggesting that the impaired bone regeneration is likely attributed to MSC dysfunction (Figure 4G). Altogether, our findings indicate that DOX-mediated defects in MSC stem cell function negatively impact both bone maintenance and fracture healing.

Lineage trajectory inference reveals impaired MSC osteogenesis

To define how DOX alters MSC lineage specification, we mapped transcription factor (TF) activity in our niche scRNA-seq data set using single-cell regulatory network inference and clustering (SCENIC) workflow.¹⁹ As expected, SCENIC analysis accurately identified lineage-specific TF activity across niche compartments (supplemental Figure 5A).^{20–22} Within MSC-Adipo and MSC-Osteo subsets, DOX exposure was associated with an increase in predicted activities of key adipogenic TFs, including Cebpa(+) and Pparg(+)²³ (Figure 5A–B). On the contrary, DOX treatment reduced Atf4(+) activity, an important driver of osteogenic differentiation.²⁴ We also found increased Ebf3(+) activity in both MSC-Adipo and MSC-Osteo subsets after DOX treatment. As EBF3 is known to support MSC and HSC niche function by repressing osteogenic differentiation,²⁵ these results point to a DOX-driven remodeling of MSC transcriptional regulatory networks that favors adipogenic programs and suppresses skeletal differentiation.

To further resolve MSC heterogeneity, we subclustered MSC-Adipo and MSC-Osteo populations independently (supplemental Figure 5B–C). MSC-Adipo included 4 subclusters, including a nondistinct MSC_A0 cluster, early adipogenic clusters (MSC_A1 and MSC_A2), and MSC_A3, an intermediate state transitioning into osteogenic programs. MSC-Osteo spanned into 4 subclusters along the skeletal lineage, including osteogenic-primed MSCs (MSC_O0 and MSC_O1), osteoprogenitors (MSC_O2), and preosteoblasts (MSC_O3). DOX-treated mice showed increased frequencies of MSC_A0, MSC_A2, and MSC_A3, together with reduced abundance of clusters at the later stage of osteogenic differentiation, MSC_O1–O3 (Figure 5C), consistent with diminished progression along the skeletal lineage.

To explore DOX-mediated changes in MSC lineage dynamics, we performed RNA velocity-based trajectory analysis with scVelo. At steady state, adipogenic-primed MSCs

(MSC_A0–A2) progressed into an intermediate state (MSC_A3), which then underwent terminal osteogenic differentiation via MSC_O0–O3 (Figure 5D), consistent with prior pseudotime analyses.¹⁶ DOX disrupted this transition from adipogenic to osteogenic state, specifically from MSC_A3 to MSC_O0 and MSC_O0 to MSC_O1, and increased the reverse trajectory from MSC_O0 to MSC_A3 (Figure 5E). DOX-induced disorientation of osteo-primed MSCs was further highlighted by inner product analysis, which quantified differences in velocity vector direction between control and DOX MSCs (Figure 5F). We next compared predicted TF activity for each MSC subcluster. Strikingly, DOX-mediated changes in Cebpa(+), Pparg(+), Atf4(+), and Ebf3(+) activity were prominent across MSC-Adipo subclusters but confined to MSC_O0 within MSC-Osteo (Figure 5G), consistent with the differentiation block identified by RNA velocity. Together, these data pinpoint DOX-induced transcriptional disruption at the transition between adipogenic-primed and osteogenic-primed MSCs, reducing progression into downstream osteogenic precursors. Moreover, such a differentiation block likely contributes to the accumulation of MSCs in the BM and sustained deficits in MSC-derived osteoblast output and skeletal integrity after DOX.

Epigenetic and transcriptional dysregulation drives impaired MSC dysfunction

To assess DOX-induced transcriptional changes in MSCs in depth, we performed bulk RNA-seq on freshly isolated Lepr^{tdT} MSCs at 8 weeks either after 4 weekly doses (DOX4X) or a single dose (DOX1X) of DOX, as bone defects were evident even after a single injection (supplemental Figure 4E–F). Principal component analysis showed a clear separation between treatment groups (Figure 6A). DGE analysis comparing DOX4X with control MSCs identified 306 differentially expressed genes (DEGs; supplemental Figure 6A). DOX-treated MSCs showed upregulation of adipogenic markers and suppression of osteogenic genes, consistent with our scRNA-seq data (Figure 6B). Gene set enrichment analysis revealed downregulation of collagen biosynthesis, osteogenesis, and bone formation pathways, along with a reduced expression of transforming growth factor β and platelet-derived growth factor signaling components (Figure 6C), indicating broad disruption of MSC lineage commitment and skeletal regulatory programs.

In contrast, DOX-upregulated pathways were related to DNA damage response and telomere maintenance. DOX-treated MSCs also showed increased expression of genes associated with chromatin instability, homologous recombination, and Fanconi anemia pathways (Figure 6D),^{27,28} indicating sustained

Figure 6 (continued) enrichment analysis (GSEA) analysis comparing DOX4X with the control. Kolmogorov-Smirnov test with FDR correction. (D) Expression of differentially expressed genes involved in DNA damage response and inflammatory response in control and DOX-treated MSCs. Wald test with FDR correction. (E) Acute induction of DNA damage following DOX (1 $\times$, 5 mg/kg) treatment. Representative flow cytometry analysis of γ H2AX staining on MSCs (left). MSCs isolated 2 hours after in vivo irradiation (5.5 Gy) were used for the positive control for γ H2AX staining. Geometric mean fluorescence intensity of γ H2AX in MSCs at 20 hours after DOX treatment, normalized to control (right). $n = 7$; unpaired t test. (F) PCA plot of MSC ATAC-seq at 8 weeks following control and serial DOX (5 mg/kg) treatment. $n = 3$. (G) Genomic regions enrichment of annotations tool analysis of differentially accessible peaks comparing DOX4X with control. Binomial test with FDR correction. (H) Representative snapshots of genomic loci of significantly altered genes associated with adipogenesis and osteogenesis. For each gene, the same intensity scaling is applied to both treatment groups. Wald test with FDR correction. (I) Schematic showing the study design of transcriptional profiling on human BM stroma. Paired BM aspirates were collected from patients with ALL at diagnosis and at 4 weeks after treatment (completion of induction therapy). BM stroma was expanded until passage 1, followed by RNA-seq. (J) PCA plot of RNA-seq data set of BM stroma obtained from patient BM aspirates. $n = 8$. (K) GSEA analysis comparing treated vs diagnosis samples. Kolmogorov-Smirnov test with FDR correction. (L) GSEA analysis of MSC signatures derived from human BM niche data set by Bandopadhyay et al²⁶ (top 50 differentially expressed markers for each MSC subset). Kolmogorov-Smirnov test with FDR correction. Adj. p , adjusted probability value; ATAC-seq, assay for transposase-accessible chromatin using sequencing; ATR, ataxia telangiectasia and rad3-related; CNTRL, control; CR, complete remission; Gy, gray; h, hours; IL, interleukin; NES, normalized enrichment score; PC, principle component; PDGF, platelet-derived growth factor; R/R, relapsed/refractory; TGFBR, transforming growth factor β receptor; THY1, thymocyte differentiation antigen 1; TP53, tumor protein p53; γ H2AX, phosphorylated H2A histone family member X. Created in BioRender.com. Li, X. (2026) <https://BioRender.com/zdi6tp4>.

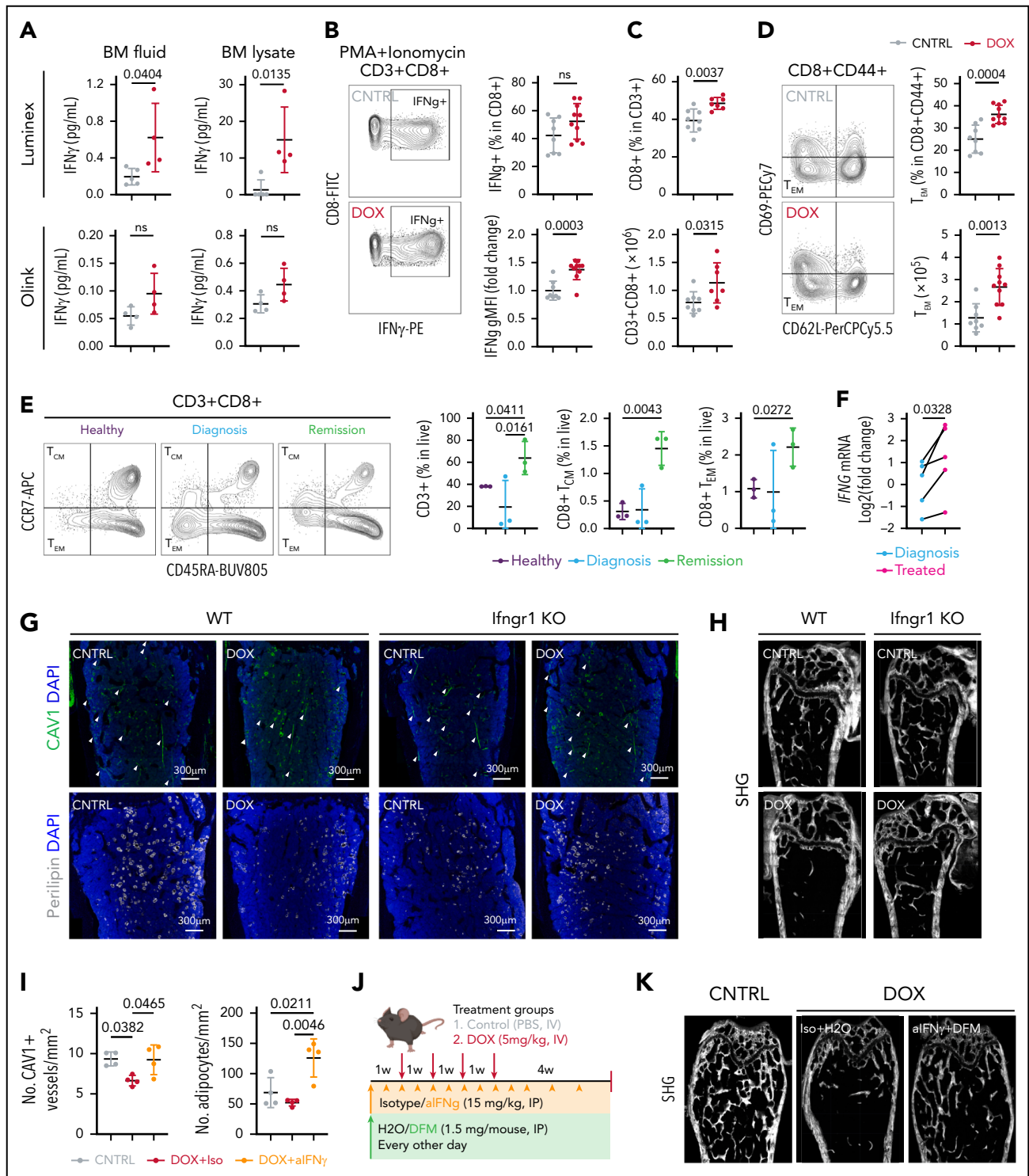


Figure 7. IFN- γ from CD8 $^{+}$ T cells drives BM niche dysfunction following DOX treatment. (A) IFN- γ level in BM lysate and BM fluid detected in Luminex and Olink multiplex cytokine assays at 8 weeks after control and serial DOX treatment. $n = 4$ to 5 ; unpaired t test. (B) IFN- γ production from CD8 $^{+}$ T cells exposed to DOX treatment upon stimulation with phorbol 12-myristate 13-acetate and ionomycin. Representative flow cytometry image (left). Proportion of IFN- γ -producing cells in CD8 $^{+}$ T cells and geometric mean fluorescence intensity of IFN- γ staining in IFN- γ -producing CD8 $^{+}$ T cells (right). $n = 8$ to 10 ; unpaired t test. (C) Relative frequency and number of BM CD8 $^{+}$ T cells. $n = 7$ to 8 ; unpaired t test. (D) Abundance of CD8 $^{+}$ T_{EM} in BM at 8 weeks after treatment. Representative flow cytometry image (left). Relative frequency and number of CD8 $^{+}$ T_{EM} in BM (right). $n = 8$ to 10 ; unpaired t test. (E) T-cell composition in BM aspirates from patients with leukemia after CTX. Representative flow cytometry analysis on memory T cells (left). Frequencies of CD3 $^{+}$ T cells, CD8 $^{+}$ T_{CM}, and CD8 $^{+}$ T_{EM} cells in the BM (right). $n = 3$; unpaired t test for healthy vs diagnosis and healthy vs remission; paired t test for diagnosis vs remission. (F) IFN γ mRNA expression in BM samples of patients with leukemia collected at diagnosis and at 4 to 12 weeks after diagnosis. $n = 5$; paired t test. (G) Representative images of CAV1 $^{+}$ arteriolar vessels and perilipin $^{+}$ adipocytes in femurs of WT and *Ifngr1* KO animals 8 weeks after DOX treatment, male. (H) Representative second harmonic generation images showing trabecular bone area in femurs of WT and *Ifngr1* KO animals after DOX treatment, male. (I) Number of CAV1 $^{+}$ arteriolar vessels and perilipin $^{+}$ adipocytes in femurs after single DOX treatment (10 mg/kg) and IFN- γ blockade (15 mg/kg), quantified based on supplemental Figure 8G. $n = 4$; 1-way analysis of variance followed by multiple comparison tests with FDR correction. (J) Experiment schematic of aIFN- γ (15 mg/kg) and DFM

genotoxic stress. Flow cytometry analysis showed γ H2AX induction at 20 hours but not at 3 days after treatment (Figure 6E; supplemental Figure 6B). Despite acute resolution of DNA damage, the persistence of DNA damage programs at 8 weeks suggests a durable cellular stress state and/or ongoing genomic instability in MSCs long after the acute insult. In addition, we identified significant upregulation of genes and pathways associated with inflammatory signaling after DOX treatment (Figure 6C-D).²⁹⁻³² Together with the increased expression of inflammatory genes detected in ECs in our scRNA-seq data (Figure 2I-J), these findings indicate that DOX induces sustained inflammation, potentially driving long-term remodeling of the BM microenvironment.

To determine whether these MSC transcriptional changes were dose dependent, we compared DOX1X and DOX4X regimens. Principal component analysis showed that DOX1X samples clustered between control and DOX4X groups, indicating a graded response (Figure 6A). DOX1X vs control group comparison revealed 167 DEGs, with only 7 DEGs between DOX1X and DOX4X (supplemental Figure 6A). Gene-level changes were strongly correlated (supplemental Figure 6C), and >70% of enriched pathways overlapped between the 2 DOX treatment groups (supplemental Figure 6D). These findings suggest that even a single DOX dose is sufficient to trigger persistent transcriptional reprogramming in MSCs, which intensifies with repeated exposure.

The persistence of MSC fate skewing after DOX suggests long-term alterations in the chromatin landscape. To this end, we performed an assay for transposase-accessible chromatin using sequencing on MSCs at 8 weeks after DOX treatment, identifying 1871 differentially accessible regions (Figure 6F; supplemental Figure 6E-F). The genomic regions enrichment of annotations tool analysis revealed increased accessibility of genomic regions associated with adipogenesis and inflammatory responses (Figure 6G-H). In contrast, regions with reduced accessibility were enriched for bone morphology gene sets. These sustained alterations in chromatin landscape at key loci indicate that changes in MSC fate dynamics are governed by epigenetic modifications.

To examine the impact of CTX on the human BM niche compartments, we expanded BM stroma from paired BM aspirates collected at ALL diagnosis and after induction of CTX and performed RNA-seq (Figure 6I-J; supplemental Figure 6G). Gene set enrichment analysis revealed down-regulation of pathways associated with protein translation, oxidative phosphorylation, and DNA replication after treatment (Figure 6K). Stromal cultures derived from BM samples collected after induction therapy also exhibited reduced growth compared with diagnostic samples, consistent with impaired proliferative capacity observed in DOX-treated murine MSCs (supplemental Figure 6H). In parallel, inflammatory signaling pathways were upregulated, along with enrichment of extracellular matrix deposition and collagen

metabolism pathways in BM stroma following CTX. We further observed increased expression of gene programs linked to adipogenesis and lipid metabolism, suggesting altered MSC lineage potential. To assess this, we leveraged a published human BM niche scRNA-seq data set²⁶ to generate MSC subset signatures and evaluate MSC lineage potential in our BM stroma data. Adipo-MSC and THY1⁺ MSC signatures, which both correlate to adipogenic-primed MSCs in mice, were enriched in postinduction BM stroma (Figure 6L; supplemental Figure 6I). In contrast, signatures associated with osteolineage-primed subsets, including Osteo-MSC and APOD⁺ MSCs, were reduced after induction therapy. Collectively, these data indicate that CTX is associated with reduced proliferative capacity, increased inflammatory signaling, and altered lineage priming in human BM stroma.

DOX treatment induces a proinflammatory microenvironment in the BM

Conserved upregulation of inflammatory pathways in MSCs and ECs after DOX treatment suggests a role for chronic inflammation in BM niche disruption. Multiplex cytokine assays detected elevated interferon gamma (IFN- γ) in BM fluid and lysate, but not in serum, at 8 weeks after treatment, indicating a tissue-localized inflammatory response (Figure 7A; supplemental Figure 7A). We next sought to identify the cellular source of IFN- γ after DOX treatment. Given that IFN- γ is primarily produced by T cells and natural killer cells, we examined their intracellular IFN- γ levels at 8 weeks after treatment (supplemental Figure 7B). Strikingly, we observed elevated IFN- γ production from CD8⁺ T cells upon in vitro stimulation (Figure 7B). Moreover, flow cytometry analysis showed a marked increase in CD8⁺ T cells in BM at 8 weeks after treatment (Figure 7C). Within the CD8⁺ T-cell compartment, we identified a significant increase in the proportion and number of CD8⁺ effector memory T (T_{EM}) cells and the number of CD8⁺ tissue-resident memory T cells in DOX-treated mice, whereas the number of central memory CD8⁺ T cells remained comparable (Figure 7D; supplemental Figure 7C-D). These results show increased activation and effector functions of CD8⁺ T cells after DOX treatment.

To examine whether similar immune remodeling is observed in patients treated with CTX, we performed immunophenotyping on healthy BM controls and paired BM aspirates collected from patients with B-cell ALL at diagnosis and in remission (supplemental Figure 7E). At diagnosis, B-cell ALL BM showed increased frequencies of CD8⁺ T_{EM} cells compared with healthy controls, consistent with T-cell activation in response to leukemia (supplemental Figure 7F). A trend toward increased CD8⁺ T_{EM} frequency was also observed in remission samples relative to controls ($P = .0634$). Importantly, we observed significant expansion of T cells, CD8⁺ T_{EM}, and CD8⁺ central memory T cells in remission BM compared with healthy controls (Figure 7E). Notably, all patients had achieved complete remission by minimal residual disease assessment at least 4 weeks before sample collection. Although both leukemia and

Figure 7 (continued) (1.5 mg per mouse) combination treatment. (K) Representative second harmonic generation images of femurs in DOX-treated mice with and without aIFN- γ and DFM treatment, male. CNTRL, control; IP, intraperitoneal; KO, knockout; PE, phycerythrin; PMA, phorbol 12-myristate 13-acetate; MFI, mean fluorescence intensity; ns, not significant; SHG, second harmonic generation; T_{CM}, central memory T cells; T_{RM}, tissue-resident memory T cell; w, weeks; WT, wild-type. Created with BioRender.com. Li, X. (2026) <https://BioRender.com/zdi6tp4>.

CTX may contribute to T-cell activation, the persistence of an expanded memory T-cell pool in remission indicates sustained immune remodeling even in the absence of detectable disease. Consistent with this, *IFNG* expression was increased in post-CTX BM compared with paired diagnostic samples (Figure 7F). Together, these findings indicate expansion of T_{EM} cells and primed IFN- γ production in patients with leukemia after CTX, similar to that in murine models.

Next, we examined whether BM niche compartments are responsive to IFN- γ signaling. Flow cytometry analysis confirmed surface expression of IFNGR1 on MSCs and ECs (supplemental Figure 8A). Reanalysis of our bulk RNA-seq data identified significant upregulation of IFN- γ signaling in DOX-treated MSCs (supplemental Figure 8B). Moreover, DOX-treated EC subsets showed increased expression of IFN- γ -responsive genes, including major histocompatibility complex class I components (*B2m*, *H2-D1*, and *H2-K1*; Figure 2I).³³ SCENIC analysis showed an increased activity of downstream TFs, such as *Irf1*(+), *Irf2*(+), and *Stat1*(+), in both MSC and EC subsets (supplemental Figure 8C), implying a sustained IFN- γ -driven inflammatory milieu after DOX treatment. Consistently, we observed an increased expression of IFN- γ -mediated responses in patient BM stroma exposed to CTX (Figure 6K; supplemental Figure 8D). To test the functional consequences of IFN- γ exposure, we assessed its direct effects on MSCs in vitro. In fibroblast colony-forming unit assays, we documented a dose-dependent reduction in MSC clonogenic potential upon IFN- γ treatment (supplemental Figure 8E). Moreover, both osteogenic and adipogenic differentiation potential were significantly impaired by IFN- γ treatment (supplemental Figure 8F).

To define the functional contribution of IFN- γ to DOX-induced BM niche remodeling in vivo, we subjected *Ifngr1* knockout mice to serial DOX treatment. Compared with wild-type controls, *Ifngr1* knockout mice retained higher arteriolar density, exhibited partial rescue of MSC adipogenic differentiation (Figure 7G; supplemental Figure 8G), and were protected from DOX-induced trabecular bone loss at 8 weeks after treatment (Figure 7H). Consistent with these findings, treatment with IFN- γ -neutralizing antibody (aIFN- γ) increased CAV1⁺ arteriolar vessels and restored MSC adipogenesis (supplemental Figure 8H-I; Figure 7I). Similarly, antibody-mediated ablation of CD8⁺ T cells mitigated DOX-mediated arteriolar vascular injury, suggesting that CD8⁺ T cells are likely the primary source of IFN- γ after DOX treatment (supplemental Figure 8J-K).

Given the critical role of arteriolar vasculature in supporting osteogenesis, we reasoned that augmenting vascular recovery might further improve bone outcomes. To test this, DOX-treated mice were cotreated with aIFN- γ and deferoxamine mesylate (DFM) (Figure 7J). Although used clinically as an iron chelator, DFM also promotes BM angiogenesis via HIF1 α stabilization¹¹ and exhibits antioxidant and immunomodulatory properties.^{34,35} Strikingly, combined aIFN- γ and DFM treatment, but not DFM or aIFN- γ alone, significantly improved trabecular bone area in DOX-treated mice (Figure 7K; supplemental Figure 8L-M). These findings indicate that DOX-induced osteotoxicity can be alleviated through coordinated suppression of inflammatory signaling and promotion of vascular recovery.

Discussion

Although many cancer survivors achieve remission, lasting dysfunction often persists in tissues that support hematopoietic recovery. Peripheral blood counts may normalize, yet the BM stem cell niche remains profoundly compromised. Previous studies using cisplatin,³⁶ busulfan,³⁷ and cyclophosphamide³⁸ have demonstrated that CTX drugs can significantly damage the BM microenvironment by inducing nerve injury, vascular leakage, and a loss of osteoblasts. Such impact can be determined by dosage, age of exposure, and drug-specific mechanisms of action. In this study, we demonstrated that DOX remodels stromal and vascular compartments, impairing BM support of HSCs. MSCs undergo differentiation arrest with suppressed osteogenic programs, whereas arteriolar ECs and pericytes are diminished, disrupting vascular regulation of HSC quiescence. Together, these changes dismantle the HSC-supportive niche and promote chronic inflammatory dysregulation.

Chronic inflammatory signaling, centered on IFN- γ , emerged as a key mediator of DOX-induced BM niche dysfunction. Previous work has shown that IFN- γ treatment impaired MSC proliferation and adipogenic differentiation in vitro.^{39,40} Moreover, mice with aberrant IFN- γ production exhibited loss of PDGFR α ⁺CD51⁺ MSCs and reduced adipocyte accumulation in the BM. Our work links CD8⁺ T-cell-derived IFN- γ specifically to long-term BM niche defects induced by CTX. We demonstrate that DOX treatment activates CD8⁺ T cells and induces IFN- γ production in the BM, which is associated with persistent transcriptional reprogramming of MSCs and ECs. Inhibition of IFN- γ signaling partially restored MSC adipogenic differentiation and vascular integrity, whereas combined IFN- γ blockade and DFM treatment protected trabecular bone, indicating that inflammatory injury and impaired vascular recovery together contribute to stromal dysfunction. CD8⁺ T cells represented the dominant source of IFN- γ in the BM after DOX treatment. However, the mechanisms driving CD8⁺ T-cell activation, including whether responses are antigen specific or represent broad immune dysregulation, warrant future investigation. The concordant inflammatory activation and MSC lineage-priming programs observed across both mouse and human data identify IFN- γ as a conserved mechanistic driver of DOX-mediated BM niche remodeling. Notably, studies in DOX-induced cardiac injury have similarly implicated aberrant CD8⁺ T-cell activation in driving tissue dysfunction,⁴¹ consistent with a more general paradigm of immune-mediated tissue injury after CTX.

Secondary hematologic malignancies are recognized as late complications of genotoxic therapy.² In the steady-state setting, the post-DOX BM niche is chronically inflamed, driving loss of HSC quiescence and phenotypic expansion of HSCs.^{42,43} Conversely, in the transplantation setting, where hematopoietic recovery places high proliferative and self-renewal demands on HSCs, the disrupted niche fails to adequately support this stress response, resulting in impaired long-term HSC maintenance. In this context, reduced niche support combined with persistent inflammatory signaling may create selective pressure favoring the expansion of preexisting or therapy-induced mutant clones. Therefore, chronic IFN- γ -driven niche dysfunction, together with diminished vascular and stromal support, may shift hematopoiesis toward maladaptive regeneration rather than restoration of normal stem

cell homeostasis. These findings raise the possibility that BM niche damage contributes to therapy-related hematologic malignancies and highlight the need to define how CTX-altered niches influence genetically predisposed HSCs.

Study limitations

B-cell populations were defined based on B220 expression, which is also present on non-B-cell populations. In addition to CD8⁺ T cells, other immune cell populations may contribute to IFN- γ production after DOX treatment. Although we demonstrated that IFN- γ blockade partially restored structural integrity of the BM microenvironment, its impact on HSC maintenance at baseline or after transplantation will be an important focus for future studies.

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Authorship

Contribution: X.L. and A.N.T. conceived and designed the study; X.L. performed most of the experiments; A.G.A.-N., M.N., M.G., M.F., D.Y., B.L., D.O., and C.K. assisted with experiments and contributed intellectual input; D.-M.P., S.B., and D.M. performed microcomputed tomography scanning and analysis; D.M. and A.C. performed 3-point bending

tests and analysis; S.J., H.S.L., G.E., P.K., and G.S. conducted computational and statistical analyses; S.F. and K.K.S. assisted with confocal imaging; C.K. performed histology analysis; M.M., A.A., J.H., K.J.L., and Y.-M.K. provided patient samples and clinical information; C.W., J.B., T.K., T.M., M.K., D.C., B.O.Z., S.N., M.A., C.J., F.G., and R.H.A. shared reagents, analytical tools, and expertise and provided intellectual input; and X.L. and A.N.T. wrote the manuscript.

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Footnotes

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The online version of this article contains a data supplement.

There is a *Blood Commentary* on this article in this issue.

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